

Interior Permanent Magnet Motor Design

Design to Improve the Formula Student Team Car's Performance

IPM Design Team

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1 Background: Current Motor Analysis

The KTH Formula Student team currently utilizes the **DD5-14-10-POW** motor to power their vehicle. This vehicle is used in different competitions with different terrains. From a flat long track where speed is needed, to auto-cross scenarios where torque is critical. Due to that reason, the car should be able to perform in a variety of conditions. The following characteristics were identified:

- **Winding Configuration:** 86 wire ends, indicating a total number of turns per tooth of 43.
- **Inductance Profile:** The baseline motor exhibits an inverse saliency where $L_d = 0.24$ mH and $L_q = 0.12$ mH ($L_q < L_d$).



Figure 1: Stator traversal cut

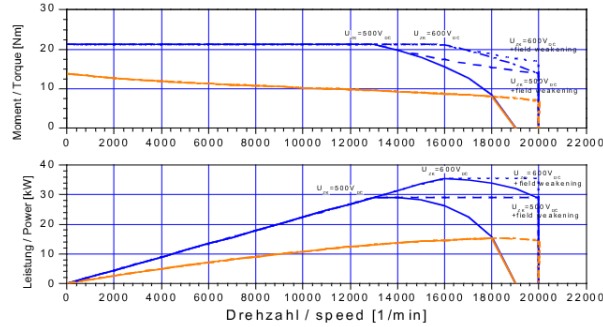


Figure 2: Power and Torque vs Current for the current motor.

We can see how the winding heads space have to be considered for the stack length calculation; meaning that the true stack length limit is $110\text{mm} - \text{winding heads} = 85.2\text{ mm}$

2 Part A: Framework Definition and Requirements

The design objective is to transition from the current baseline to an IPM topology while maintaining fixed outer dimensions to ensure compatibility with the existing chassis.

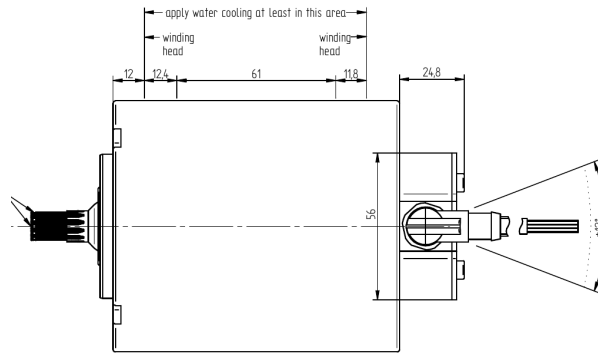


Figure 3: Length of the current motor.

2.1 Boundary Requirements

The following table summarizes the design boundaries, which are defined by the inverter datasheet, current design and the Formula Team itself. The new design should stick to these requirements, and any design deviation should be justified.

Table 1: Specifications and requirements for the new propulsion motor design.

General requirements		
Max dimensions including cooling (outer diameter $\times$ length)	120 mm $\times$ 110 mm	
Aspect ratio (length/diameter)	0.9167	
Hub compatible	Must	
Motor topology	Radial in-runner (rotor inside the stator)	
Weight	≤ 2800 g	
Power density (mass)	≥ 10 kW/kg	
IP protection	IP65	
Mechanical requirements		
Peak torque (for 5 s)	21.5 Nm	
Maximum speed (operating cap)	12000 rpm	
Nominal torque	9.8 Nm	
Nominal speed	12000 rpm	
Electrical requirements		
Source Voltage (input for inverter)	515 V_{DC}	
Machine Voltage Input ($V_{LL,RMS}$)	334 V_{AC}	
Machine Voltage Input ($V_{LN,RMS}$)	202 V_{AC}	
Peak power	28 kW	
Average power	12 kW	
Three-phase cable diameter	11 mm $\pm$ 0.5 mm	
Sensor requirements		
Rotor position	Must	
Stator temperature	Must	
Rotor temperature	Preferred	
Cooling requirements		
Coolant	Liquid or air	
Cooling circuit	External or internal	
Compatibility		
Rotary encoder	ECI1118, P-type	
Output spline	DIN 5480	-
	W11x0,8x30x12x7h	

2.2 First Design Objectives

For the new design, we will choose to use permanent magnets to achieve the required power density, making sure to take into consideration the thermal limits of the magnets. A surface permanent magnet motor (SPM), although simple and cheap, would need a very good glue in order to be able to withstand the centrifugal forces at the operating cap of 12,000 rpm. Because of those reasons, we decide to proceed with an IPM Machine design. Increasing the winding turns from the current 43 will allow for higher magnetic flux linkage and thus letting the back-EMF reach the DC bus voltage limit at a lower speed, while reducing I^2R losses for the nominal speed. Also, this will enable us to use the motor in the field-weakening mode for a wider speed range. Finally, changing the L_d and L_q proportions (letting $L_q > L_d$) will allow us to operate in the field-weakening region by using d-axis current injection and counteract the permanent

magnet flux, maintaining a flatter power curve.

Number Of Slots And Poles

Using standard designs as a first reference [??], we select a **12-slot / 10-pole** configuration (i.e., $p = 5$ pole pairs)

The corresponding electrical frequency is

$$f_{el} = \frac{p \cdot n}{60} = \frac{5 \cdot 12,000}{60} = 1000 \text{ Hz} \quad (1)$$

At the real operating cap of $n_{max} = 12,000$ rpm.

Theoretical Foundation of Base Speed

The point where torque begins to fall is defined as the Base Speed (ω_{base}), occurring when the motor's internal Back-EMF (Back-EMF) reaches the maximum voltage available from the inverter. This relationship is governed by:

$$\omega_{m,base} \approx \frac{V_{\text{phase,peak,max}}}{p \Psi_m} \quad (2)$$

Where $V_{\text{phase,peak,max}}$ is the maximum achievable *phase* fundamental peak voltage, p is the number of pole pairs, and Ψ_m is the permanent magnet flux linkage.

The currently used AMK inverter considers a maximum continuous modulation index of $M_i = 0.917$, which limits the maximum fundamental phase voltage to:

$$V_{\text{phase,peak,max}} = M_i \frac{V_{DC}}{\sqrt{3}} \approx 0.917 \times \frac{515}{\sqrt{3}} \approx 273 \text{ V} \quad (3)$$

To trigger the torque droop at an earlier speed, we must increase Ψ_m (e.g., by increasing stator turns or selecting higher remanence magnets), thereby reducing $\omega_{m,base}$.

For example, targeting $n_{base} = 10,000$ rpm with $p = 5$ gives a mechanical speed of $\omega_{m,base} \approx 1047$ rad/s. Using the available $V_{\text{phase,peak,max}} \approx 273$ V yields a required order-of-magnitude flux linkage of:

$$\Psi_m \approx \frac{V_{\text{phase,peak,max}}}{p \omega_{m,base}} \approx \frac{273}{5 \cdot 1047} \approx 0.0528 \text{ Wb} \quad (4)$$

Analytical Sizing using Essen's Rule

To translate the performance requirements into initial physical dimensions, we utilize **Essen's Rule**. This fundamental sizing equation relates the required peak torque to the active volume of the motor's rotor.

The Sizing Equation

The relationship between torque and rotor volume is defined as:

$$T = C_0 \cdot D^2 L \quad (5)$$

Where:

- T is the peak torque target (7 Nm).
- D is the rotor outer diameter (m).
- L is the active stack length (m).
- C_0 is the Output Coefficient (or Essen's Coefficient).

Selection of the Output Coefficient (C_0)

In high-performance electric machine design, C_0 is often interpreted as the **Magnetic Shear Stress** (σ) or **Torque per Rotor Volume (TRV)**, yielding SI units of N/m^2 .

Permanent magnet machines which do not have a rotor current disadvantage typically fall in the range from 40,000 to 60,000 N/m^2 . As an upper limit, liquid-cooled machines having a shear stress of up to 100,000 N/m^2 have been reported (Lipo, 2017). For the KTH design, we target an initial value of **50,000 N/m^2** . We can allow this design choice thanks to the cooling loop, and the fact that transient nature of racing allows for higher transient "hardness" in material loading compared to continuous-duty industrial applications.

2.3 Rotor Initial Dimensional Calculation

By rearranging the sizing equation, we calculate the required stack length (L) needed to achieve 7 Nm of torque within the 80 mm rotor diameter ($\varnothing 80$ mm) derived from the baseline AMK motor envelope:

$$L = \frac{T_{peak}}{D^2 \cdot C_0} = \frac{7 \text{ Nm}}{(0.080 \text{ m})^2 \cdot 50,000 \text{ N/m}^2} \approx 0.0175 \text{ m} \quad (6)$$

This analytical result provides the baseline geometry for subsequent high-fidelity Finite Element Analysis (FEA) in COMSOL.

The following table summarizes the first design objectives.

Table 2: New Design Objectives

<i>Parameter</i>		<i>Symbol</i>	<i>Value</i>	<i>Engineering Justification</i>
Pole Number		P	10	Sized to maximize the AMK RACING KIT inverter capacity (26 kVA/channel) while respecting the inverter DC bus ceiling (515 V).
Line Voltage		V_{line}	515 V	It is under the 540V nominal value of the inverter.
Target Power	Nominal	P_{nom}	10 kW	This power reduction will allow for better energy efficiency.
Target Power	Peak	P_{peak}	28 kW	Sized to maximize the AMK RACING KIT inverter capacity (26 kVA/channel) while respecting the inverter DC bus ceiling (515 V).
Target Torque	Nominal	T_{nom}	7 Nm	N/A
Clipped Torque	Peak	M_{clip}	18 Nm	Software clip to ensure mechanical reliability of the transmission and manage thermal saturation.
Target Max Speed		n_{max}	12,000 rpm	Constrained by the 1200 Hz inverter limit; provides a 20% safety margin (1000 Hz at 12,000 rpm for 10 poles).
Target Speed*	Nominal	n_{nom}	10,000 rpm	*NOTE: This value is under the 12,000 rpm constraint defined earlier. The reason for this deviation is that we are going to optimize for Field Weakening Mode.
Gear Ratio		G	14:1	Fixed by the current powertrain package to optimize acceleration out of tight autocross corners.
Rotor Topology		—	IPM	Selected to achieve $L_q > L_d$, enabling reluctance torque to hit the 18 Nm target under the 107 A inverter limit.

3 Phase 2: Analytical Design and Rotor Parameterization

To establish a robust first-principle model prior to Finite Element Analysis (FEA) in COMSOL, the analytical procedure proposed by Di Gerlando and Ricca is implemented. This approach defines the critical V-shape geometry based on structural constraints and evaluates the cross-coupling magnetic saturation.

For traceability, Table 3 summarizes the **current** geometry and operating-point parameters pushed from MATLAB into the COMSOL model.

Table 3: COMSOL parameter set used by the MATLAB automation (single source: MATLAB_to_COMSOL_Interface.m).

Parameter	Symbol	Value
Number of poles	N_p	10
Number of slots	N_s	12
Rotor diameter	d_r	80 mm
Stator diameter	d_{st}	100 mm
Airgap	g	0.7 mm
Stack length (sized)	L	from Essen's rule + winding head
Inverter current limit	I_{pk}	107 A (phase peak)
Speed (operating cap)	n_{max}	12,000 rpm
Magnet width (slab)	w_m	14 mm
Magnet thickness	t_m	3.5 mm
V-angle	α_v	130°
Outer bridge thickness	b_t	1.0 mm

3.1 Rotor Bridge Sizing and Structural Integrity

The structural integrity of the IPM rotor is heavily dependent on the sizing of the inner (w_{ib}) and outer (w_{ob}) iron bridges. In this project the real operating cap is $n_{max} = 12,000$ rpm (inverter/electrical frequency constraint), and the bridge stress check is performed at this speed. Electromagnetically, these bridges must be as thin as possible to minimize permanent magnet (PM) flux leakage. Mechanically, they must withstand the centrifugal forces exerted by the PM segments and the iron pole shoe.

Following the analytical model, the outer bridge is fixed to a minimum practical manufacturing thickness. In the current COMSOL iteration this is set to $b_t = 1.0$ mm, while the inner bridge is sized to sustain the primary centrifugal load.

The maximum angular velocity for a mechanical speed n_{max} is:

$$\Omega_{max} = \frac{2\pi n_{max}}{60} \quad (7)$$

For $n_{max} = 12,000$ rpm, $\Omega_{max} \approx 1257$ rad/s.

The specific maximum centrifugal force (f_{max}) is calculated based on the mass of the pole shoe and magnets (m_{1p}) at average radius (R_{av}):

$$f_{max} = m_{1p} \cdot R_{av} \cdot \Omega_{max}^2 \quad (8)$$

As a first-order estimate, the magnet contribution to m_{1p} can be computed from the present geometry (two magnets per pole) as:

$$m_{pm,1p} \approx 2 \rho_{pm} w_m t_m L \quad (9)$$

where ρ_{pm} is the magnet density, $w_m = 14$ mm, $t_m = 3.5$ mm, and L is the active stack length.

The stress on the inner bridge is then evaluated against the lamination yield strength ($\sigma_{y,lam}$) using a stress concentration factor (K_t):

$$\sigma_{ib} = K_t \cdot \frac{f_{max}}{w_{ib} \cdot k_{st}} \leq \sigma_{y,lam} \quad (10)$$

3.2 V-Shape Parametric Definition

With the 10-pole arrangement establishing a pole pitch of $\tau = 36^\circ$, the V-shape cavity is parametrically defined. The geometry is driven by the magnet width (b_m), magnet thickness (h_m), and the V-angle (ν).

In the present design iteration, these correspond to the COMSOL-controlled parameters $w_m = 14$ mm, $t_m = 3.5$ mm, and $\alpha_v = 130^\circ$.

The position of the magnets dictates the pole shoe extension (b_{ps}) and the depth of the rotor ribs (h_{ry}). To ensure proper flux focusing and prevent the d-axis flux from short-circuiting, the inner rotor diameter (D_{ir}) is defined by the depth of the pole shoe (d_{ps}) and the inner bridge height (h_{ib}):

$$D_{ir} = D_r - 2 \cdot (d_{ps} + h_{ib} + h_{ry}) \quad (11)$$

This parameterization allows the MATLAB automation script to iterate through V-angles and magnet widths without generating invalid geometries in COMSOL.

3.3 Magnetic Saturation and Optimal Phase Advance

Due to the highly saturated nature of the V-shape IPM design at the peak 107 A inverter limit, the d-q axis inductances cannot be treated as constant. The procedure utilizes a q-axis saturation model (σ_{sM}) and a PM flux cross-coupling saturation factor ($\eta_{\phi M}$) to account for the reduction of fundamental PM flux under heavy q-axis reaction current (M_{pq}).

The total electromagnetic torque is expressed as the sum of the alignment and anisotropy components:

$$T(\Delta, \gamma) = T_{al}(\Delta, \gamma) + T_{an}(\Delta, \gamma) \quad (12)$$

To maximize torque output within the current limit, the optimal phase advance angle (γ_{opt}) is dynamically calculated as a function of the linear current density (Δ). This analytical γ_{opt} provides the initial current angle for the FEA magnetostatic spatial sweep, ensuring the COMSOL model evaluates the motor at its Maximum Torque Per Ampere (MTPA) operating point.

4 Part B: Design Methodology and Validation

To validate the Phase 1 sizing dimensions, a Phase 2 FEA-based validation loop has been established using a COMSOL-MATLAB interface.

4.1 Electromechanical Power Modeling

Since the motor is not 100% efficient, we represent the electrical input power (P_{elec}) relative to the mechanical shaft power (P_{mech}) through the overall efficiency (η).

$$P_{elec} = \frac{P_{mech}}{\eta} \quad (13)$$

The electrical power for a 3-phase system is defined as:

$$P_{elec} = \frac{3}{2} V_{peak} I_{peak} \cos(\phi) \quad (14)$$

Here, V_{peak} and I_{peak} refer to *phase* sinusoidal peak quantities. Using SVPWM, a common approximation for the maximum achievable fundamental phase voltage is $V_{peak} \approx V_{dc}/\sqrt{3}$.

Rearranging for the current limit validation:

$$I_{peak} = \frac{2 \cdot P_{mech}}{\sqrt{3} \cdot V_{dc} \cdot \eta \cdot \cos(\phi)} \quad (15)$$

Using the peak power target $P_{mech} = 28$ kW with $V_{dc} = 420$ V, $\eta = 0.92$ and $\cos(\phi) = 0.90$ gives $I_{peak} \approx 93$ A, which is within the 107 A peak phase current limit used in the FEA workflow.

4.2 Simulation Framework

The simulation utilizes an automated script (`MATLAB_to_COMSOL_Interface.m`) to iterate through geometry parameters. This "Automation Engine" performs the following:

1. **Handshake:** Initializes the connection to the COMSOL Multiphysics Server.
2. **Parameter Injection:** Pushes the design and operating-point parameters (Table 3) into the COMSOL model's global parameter table.
3. **Solve:** Runs the electromagnetic study (2D magnetic FEA) for the selected operating point.
4. **Verification:** Extracts torque from a result table, computes average torque and torque ripple, and stores the results (`Simulation_Results.mat`) for comparison across design candidates.

5 Thermal and Mechanical Analysis

A critical design constraint is the existing **4 l/min liquid cooling loop**. The IPM transition allows for potentially higher power density (28 kW vs 24 kW), but the FEA results must be used to calculate I^2R and core losses to ensure the temperature does not reach the demagnetization point of the magnets or the insulation limit of the windings.

6 Conclusion

TBD